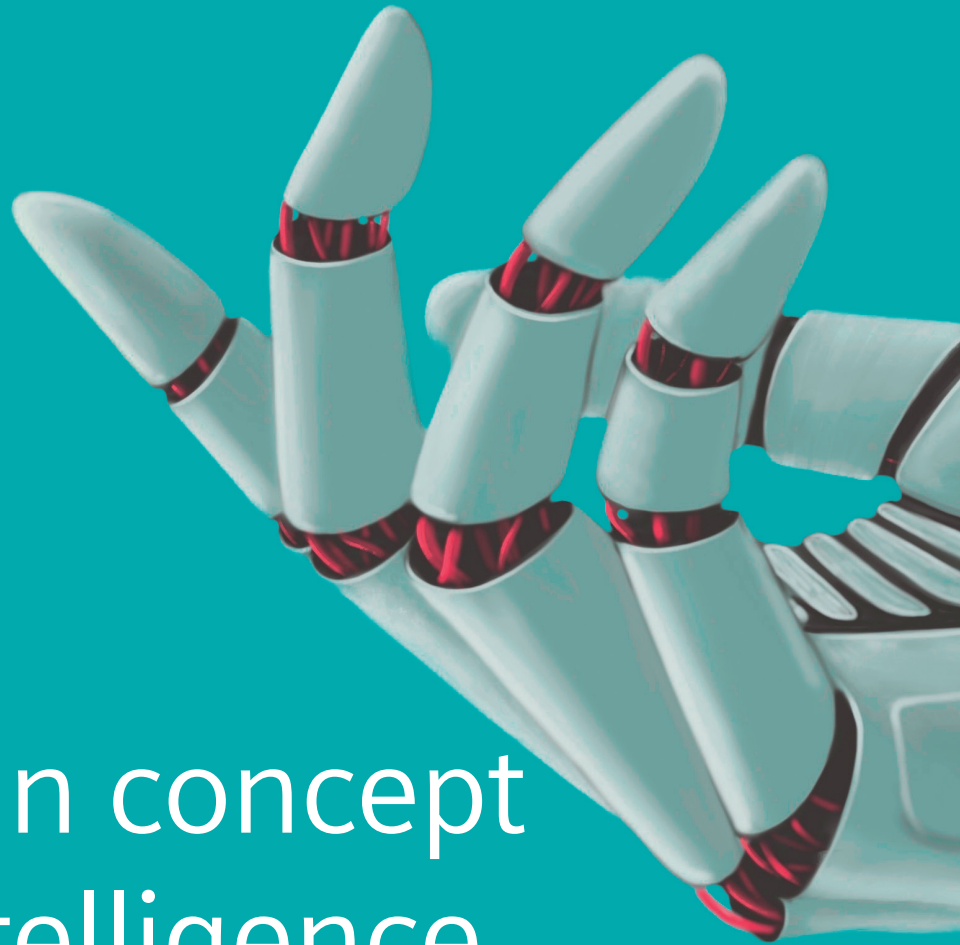


SIEMENS



DIGITAL INDUSTRIES SOFTWARE

From design concept to robot intelligence

How Designcenter enables one-time-right humanoid design

[siemens.com/software](https://www.siemens.com/software)

Executive Summary

Humanoid robots have entered a pivotal phase, transitioning from research prototypes to scalable commercial products. While advances in artificial intelligence (AI) and control algorithms dominate attention, a less visible but equally critical challenge continues to slow deployment: the disconnect between mechanical design intent and the robot intelligence platforms that depend on it.

Designcenter, the mechanical intelligence behind the Siemens

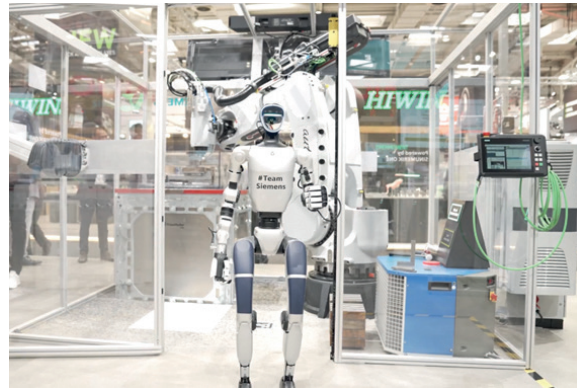
Humanoid workflow, addresses this gap by providing an end-to-end humanoid design process, from conceptual structure and kinematics definition through native Unified Robot Description Format (URDF) export, within a single, multidisciplinary environment. By preserving mechanical fidelity all the way into physics-based simulation and AI training platforms such as NVIDIA Isaac Sim, Designcenter enables a one-time-right approach not only to design, but also to behavior, motion and intelligence.



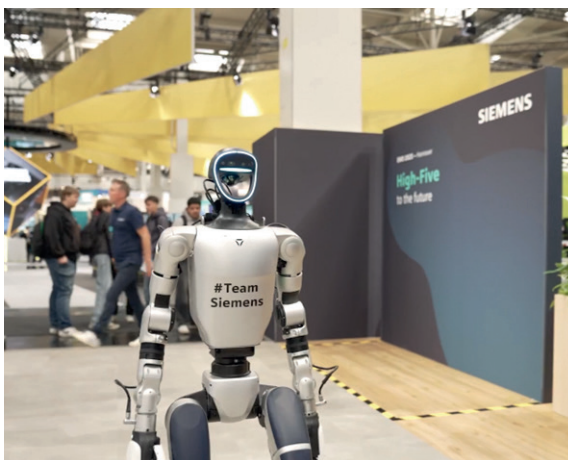
A commercial inflection point with humanoid robots

Factory floors, machine shops and all-encompassing manufacturing environments have long relied on traditional robotics to automate repetitive processes and expedite the creation of mass consumer goods. Yet the biggest challenge posed by traditional robots is adaptability: non-humanoid robots are typically engineered with a very specific manufacturing purpose in mind and they fulfill this purpose ad-nauseam; they cannot instantaneously adapt to the changing dynamics of the work environment without costly and time-consuming reconfiguration. This inhibition can stunt manufacturers who must often pivot to satisfy fluctuating market demands, as making one version of a product is no longer enough to survive in today's competitive landscape.

To challenge this status quo and drive greater agility into manufacturing, more and more organizations are turning toward humanoid robots as a potential



solution. Unlike traditional industrial automation, humanoids are designed around the human form, allowing them to operate in spaces, workflows and facilities originally built for people. In turn, they can readily adapt from addressing one manufacturing need to the next, driving continuous automation into factory procedures.



Beyond physical compatibility, humanoid robots introduce a fundamentally different economic and operational model. They can perform repetitive or physically demanding tasks continuously without fatigue, breaks, or risk of injury, helping address chronic labor shortages while reducing the impact of absenteeism, workplace injuries and ergonomic strain. Over the long term, this contributes to lower operational variability and can reduce costs associated with workforce turnover, training, health and safety programs, particularly for tasks that are difficult to fully automate with fixed or single-purpose robotic systems.

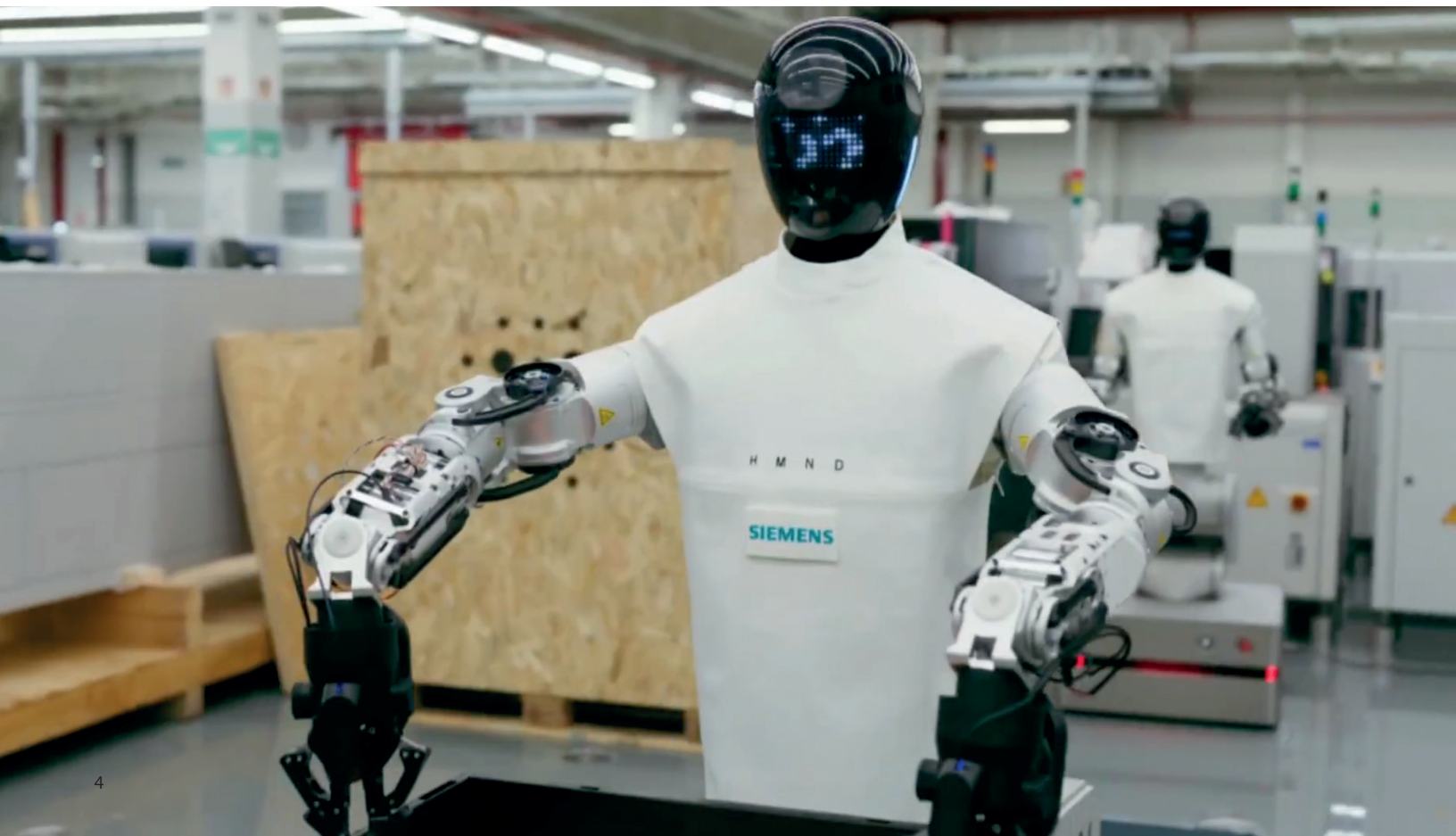
Design complexity and hidden bottlenecks

Despite the clear benefits that humanoids pose, they also introduce unprecedented design complexity. Dozens of articulated joints must work together with precise mass, inertia and center-of-gravity control. Mechanical design, electronics and control software are tightly coupled and even small changes in structure can significantly affect balance, gait, reach and energy efficiency. As a result, humanoid development has become a systems engineering challenge that demands close collaboration across mechanical, mechatronic, control and AI disciplines.

While most organizations have mature computer-aided design (CAD) processes, many humanoid programs still rely on manual and fragmented methods to translate mechanical designs into robotic simulation and control environments.

Kinematic models are often rebuilt by hand, mass and inertia properties are approximated rather than derived from real geometry and joint definitions can drift between design and control representations.

These disconnects slow iteration, undermine simulation accuracy and frequently force mechanical designs to freeze prematurely to avoid downstream rework. For humanoids, where balance, stability and manipulation are highly sensitive to physical parameters, such inaccuracies directly degrade AI training efficiency and simulation-to-reality performance. Furthermore, these design implications have direct consequences in the simulation of humanoid models, where inaccuracies impact expected performance in the virtual world, or even impact costly fixes in the physical world during physical prototyping.



Humanoid development and capabilities to consider

To create viable, fully functioning humanoids for the factory, shop floor and all-encompassing manufacturing environment, establishments need a trusted solution that eliminates systemic design obstacles and shifts the workplace mindset from consecutive to parallel product development, helping to produce humanoids at an accelerated rate without an increase in cost nor a decrease in quality.

Furthermore, to pick the right platform, organizations should evaluate whether their options possess the following capabilities:

Native URDF support, where URDF models can be exported from a CAD platform with a single click, all while mapping links, joints and kinematic hierarchies. URDF models are then compatible with PhysX-based simulation engines, preserving mechanical and kinematic fidelity from concept through robot intelligence development.

Design-simulation interoperability, as URDF models map directly into advanced simulation platforms without reinterpretation. This integration enables faster convergence between simulation and real-world behavior and supports earlier validation of balance, gait and reachability, reducing reliance on costly physical prototyping cycles.

Top-down, parametric system modeling, where master sketches and skeleton models enable rapid structural iteration without breaking downstream kinematic, motion, or simulation relationships. This directly addresses the challenge of frozen mechanical designs caused by late-stage rework.

Integrated mechanical, electrical and control co-design, driving tighter collaboration across disciplines and eliminating disconnected handoffs between teams. By keeping mechanical and control intent tightly aligned, departments can avoid misaligned joint definitions and keep models consistent between design and control environments.

Simulation-driven design with structural, multibody and thermal analysis, allowing for performance, stability and motion behavior to be evaluated early using physically accurate data. This reduces reliance on approximated mass and inertia properties that undermine simulation fidelity.

Model-Based Definition (MBD) as a single source of truth, ensuring that geometry, kinematics, mass properties and annotations remain consistent across engineering and downstream simulation workflows. This minimizes manual reconstruction and data translation errors.

Immersive augmented and virtual reality design reviews, enabling cross-functional teams to validate structure, articulation and accessibility earlier in the humanoid development process, thereby reducing late-stage discovery of integration issues that would otherwise propagate into simulation and control models.

Real-time geometric definitions where kinematic systems' criteria are computed automatically, thereby providing direct influence over outputs like stability and performance. This approach ensures that kinematic definitions evolve in parallel with mechanical design, thus maintaining consistency throughout the development lifecycle.

Designcenter: Next-generation mechanical intelligence for humanoids

For a trusted solution that encompasses all of the aforementioned capabilities and leverages decades of engineering experience, look no further than Siemens' Designcenter. As a core component of the Siemens Xcelerator portfolio, Designcenter provides a single, integrated environment for humanoid robot development. It combines computer-aided design (CAD), computer-aided engineering (CAE), computer-aided manufacturing (CAM), motion simulation and mechatronic design into one connected system, embodying three key principles:

- **Design with the digital genesis:** As implied by the name, humanoid robotic design is setting the dawn of a new era for industrial manufacturing and like with any new paradigm, there will be several challenges to consider, from managing evolving product requirements, to supporting product evolution, to attracting the best skills within the market. Designcenter is positioned to support rapid design changes and product iterations through the cloud, making it easier than ever to scale humanoid development remotely and globally so as to maintain expert talent all over, all while integrating with other solutions like product life-cycle management (PLM) that make it easier than ever to account for requirements and bill of materials (BOM)-level changes.
- **Realize with creation flow:** As organizations shift from design into prototyping development, having the tools to support production scale-up, keep costs low and maintain product quality will be paramount. Designcenter makes it possible to validate designs against the needs of mass production and automation through its integration with simulation solutions. Furthermore, by

helping establishments manage the sourcing of specialized, mechanical, electrical and software components, businesses have direct visibility into costs and can manage these facets accordingly, particularly as they begin increasing order size to meet market demands. This decrease in cost cannot come at the sacrifice of quality and so Designcenter supports quality system integrations to create accountability and support safety.

- **Operate and optimize with the coexistence loop:** Finally, as humanoid robots gain mass adoption within workplace environments, there will be an anticipated learning curve as these machines adjust and calibrate throughout deployment. Humanoids will operate alongside humans, using real-time sensing, monitoring and adaption to coordinate behavior, speed and force so as to perpetuate a secure and safe working environment. This co-existence loop will connect execution, feedback, learning and re-deployment, turning every task and action into input for improved performance, reliability and autonomy. Designcenter supports this phase of humanoid development by using data from in-field robots to drive the design and development of future humanoid iterations, making it possible to optimize and scale products writ large.

Under these three-pronged unified approach, Designcenter provides widescale coverage over humanoid development and functions as the mechanical intelligence layer within humanoid development workflows, ensuring that design intent, physical behavior and downstream simulation remain continuously aligned.

Conclusion: From One-Time-Right Design to One-Time-Right Intelligence

Traditional development processes treat mechanical design and robot intelligence as sequential activities, making late-stage changes costly and unpredictable. Furthermore, as humanoid robots scale toward commercial deployment, success depends not only on advanced algorithms, but also on the fidelity of the mechanical data driving those algorithms. Designcenter enables continuous iteration between geometry, kinematics and motion, supporting parallel collaboration across mechanical, control and AI teams, all while driving data integrity and consistency across parameters like joints, motions and hierarchies.

By maintaining a single authoritative model, from structure through URDF, designs can evolve consistently from concept through operation.

Consequently, teams can move beyond “design it right once” toward “design it right, train it right and deploy it right, always.”

In summary, Designcenter is more than a design tool; it’s a foundational platform for embodied intelligence. By connecting mechanical intent directly to robot behavior, Designcenter enables organizations to reduce risk, accelerate innovation and bring practical humanoid robots to market faster and with greater confidence.

Discover how Designcenter enables one-time-right mechanical intelligence for humanoid robots. To learn more or discuss your specific requirements, contact your Siemens representative or visit siemens.com/designcenter.



Siemens Digital Industries Software helps organizations of all sizes digitally transform using software, hardware and services from the Siemens Xcelerator business platform. Siemens' software and the comprehensive Digital Twin enable companies to optimize their design, engineering and manufacturing processes to turn today's ideas into the sustainable products of the future. From chips to entire systems, from product to process, across all industries, [Siemens Digital Industries Software](#) – Accelerating transformation.

Americas (USA): 1-800-498-5351

EMEA (United Kingdom): 0800-279-0464

APAC (India): 1-800-202-6796

For additional numbers, click [here](#).